THE UNIVERSITY OF AZAD JAMMU AND KASHMIR

LAB REPORT NO. 02

SOFTWARE CONSTRUCTION & DEVELOPMENT (LAB)

COURSE TITLE	SOFTWARE CONSTRUCTION & DEVELOPMENT (LAB)
SEMESTER	Fall 2026
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DEPARTMENT	Software Engineering

DEPARTMENT OF SOFTWARE ENGINEERING

Lab 2: UML Behavioural Diagrams

This lab presents three UML behavioural models: an activity diagram with swimlanes for an online food-ordering process, a state machine for a toaster, and a state machine for an elevator/lift. The diagrams show the main states or activities and the events that move the system from one step to another.

Q1: Activity Diagram (Swimlane): foodPanda Online Food Order

Introduction

An activity diagram shows the flow of activities in a process. In this swimlane diagram, the foodPanda ordering process is divided into four lanes: Customer, foodPanda App (System), Restaurant, and Rider. Each lane contains the activities performed by its participant.

➢ Swimlanes

- **Customer:** Places the order, makes payment, receives food, and rates the service.
- **foodPanda App (System):** Displays menus, processes payment, manages orders, and assigns a rider.
- **Restaurant:** Receives the order, accepts or rejects it, and prepares the food.
- **Rider:** Picks up the prepared food and delivers it to the customer.

➢ Process Explanation

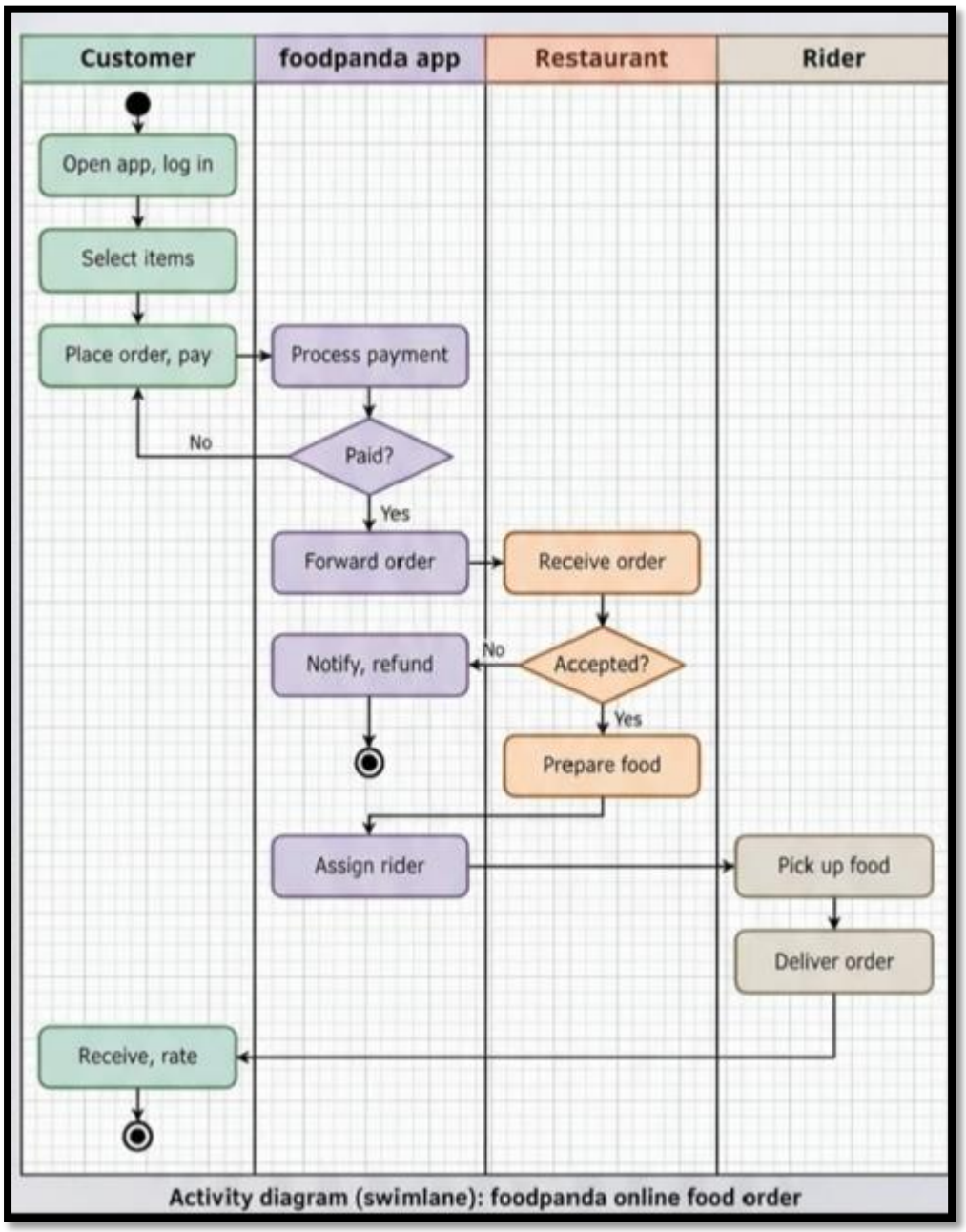
1. The process starts when the customer opens the app and logs in.
2. The customer browses restaurants and selects food items.
3. The app displays available menus and prices.
4. The customer places the order and makes payment.
5. The app checks whether payment is successful.
6. If payment fails (**No**), the customer returns to the payment step and tries again.
7. If payment succeeds (**Yes**), the app forwards the order to the restaurant.
8. The restaurant receives the order and decides whether to accept it.
9. If the restaurant rejects the order (**No**), the app notifies the customer and refunds the payment. The process ends.
10. If the restaurant accepts the order (**Yes**), it prepares the food.
11. When the food is ready, the app assigns a rider.
12. The rider picks up the food and delivers it to the customer.
13. The customer receives the food and rates the service. The process ends.

Diagram Elements

- **Filled black circle:** Start node.
- **Rounded rectangles:** Activities.
- **Diamonds:** Decision nodes (Paid? and Accepted?).

- **Bullseye circle:** End node.
- **Arrows:** Control flow between activities.
- **Swimlanes:** Show which participant is responsible for each activity.

Diagram



Q2. State Diagram of a Toaster

A state diagram shows the different states an object goes through during its life and the events that cause transitions between states. In this diagram, the object is a toaster. Each state includes three activity labels: entry, do, and exit.

States and activities

State	Activity	Example event / transition
Idle / Ready	Wait for bread and user input.	Bread inserted and lever pressed → Toasting
Toasting	Energize heating elements and monitor the timer.	Timer expires → Pop-up / Eject
Pop-up / Eject	Release toast and return lever upward.	Toast pops up → Complete
Complete	Wait for toast to be removed.	Toast removed → Idle / Ready
Cancelled / Off	Stop heating and remain inactive.	Power/start command → Idle / Ready
Fault / Overheat	Cut power and wait for safe reset.	Cool-down and reset → Idle / Ready

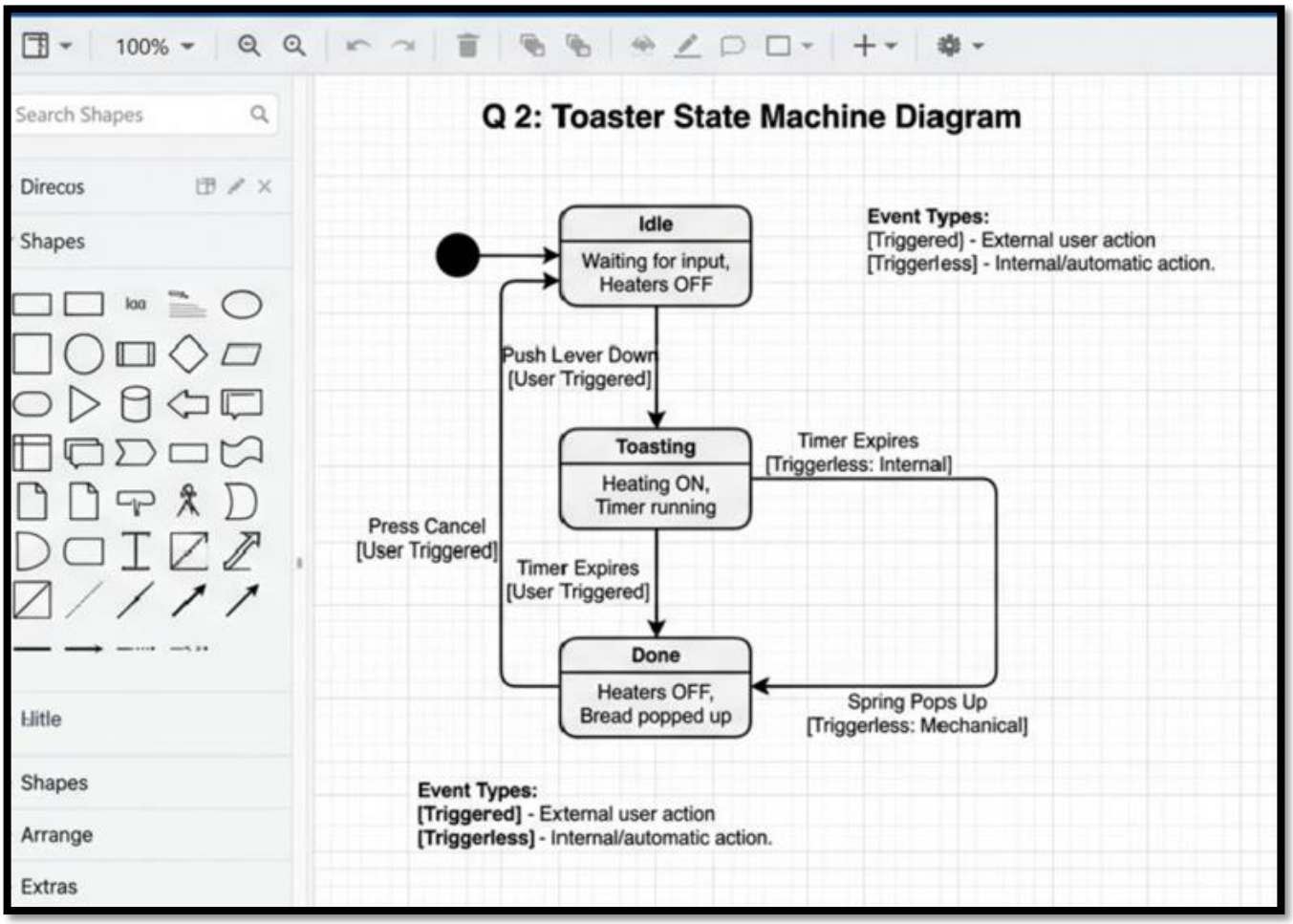
Process Explanation

1. The toaster starts in the Idle state, with the heater off and lever up.
2. When bread is inserted, the toaster moves to Bread Loaded.
3. The user selects the browning level and lowers the lever to begin toasting.
4. In the Toasting state, the heating elements operate and the timer counts down.
5. When the timer expires or the user presses cancel, the toaster moves to Ejecting.
6. The spring pushes the toast upward. Once this activity completes, the toaster automatically moves to Toast Ready.
7. When the user removes the toast, the toaster returns to Idle.
8. If the power is unplugged while in Idle, the toaster moves to the Final State.

Triggered vs. Triggerless Transitions

- **Triggered:** Caused by an identifiable event, such as inserting bread, lowering the lever, or pressing cancel. These are shown with solid arrows.
- **Triggerless:** Occur automatically when a state activity completes, without a new external event. These are shown with dashed arrows in this diagram.

Diagram:


100%
Search Shapes
Direcos
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Arrange
Extras
Q 2: Toaster State Machine Diagram
Idle
Waiting for input,
Heaters OFF
Event Types:
[Triggered] - External user action
[Triggerless] - Internal/automatic action.
Push Lever Down
[User Triggered]
Toasting
Heating ON,
Timer running
Timer Expires
[Triggerless: Internal]
Press Cancel
[User Triggered]
Timer Expires
[User Triggered]
Done
Heaters OFF,
Bread popped up
Spring Pops Up
[Triggerless: Mechanical]
Event Types:
[Triggered] - External user action
[Triggerless] - Internal/automatic action.

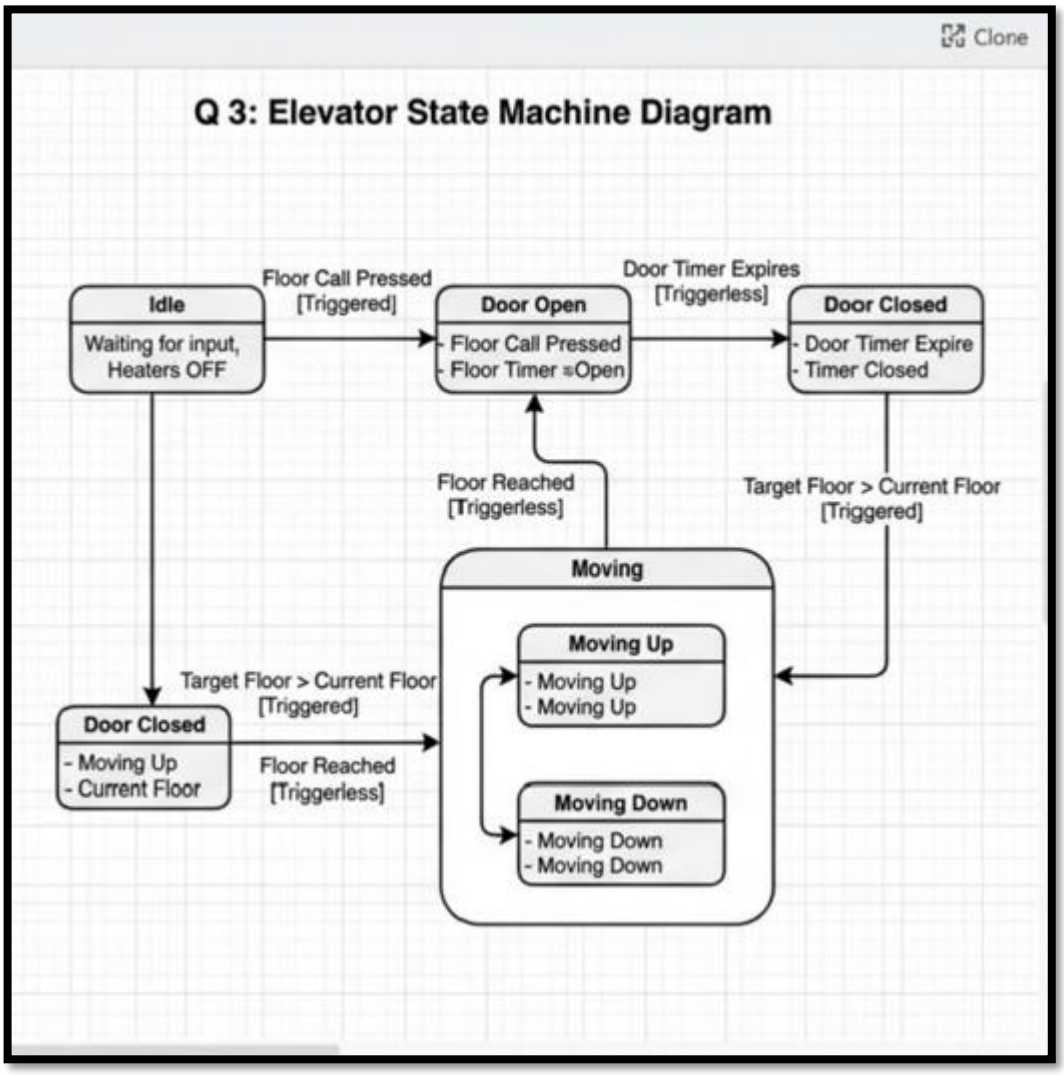
Q3. State Machine Diagram of an Elevator / Lift

Processing

The processing of the elevator state machine consists of the following steps:

1. **Initialization:** When the elevator is powered on, it enters the Idle state and waits for user requests.
2. **Request Detection:** The system checks the requested floor and determines whether the elevator should move up or down.
3. **Movement Processing:** The elevator enters Moving Up or Moving Down. The motor operates, and the floor display updates as the elevator moves.
4. **Floor Detection:** When the destination floor is reached, the floor sensor sends a signal to stop the elevator.
5. **Door Operation:** The elevator enters the Door Open state. The doors open and remain open for five seconds before closing automatically.
6. **Return to Idle:** After the doors close, the elevator returns to the Idle state and waits for another request.
7. **Emergency Handling:** If an emergency occurs during movement, the elevator enters the Emergency Stop state, applies the brake, activates the alarm, and requests maintenance.
8. **System Reset:** After the issue is resolved, an authorized technician resets the system, allowing the elevator to return to the Idle state.

Diagram:



Conclusion

The elevator state machine diagram represents the main operating states of an elevator and the transitions between them. It explains how the elevator processes floor requests, controls movement and doors, and responds to emergencies. This model helps in understanding the elevator's behavior and designing a reliable control system.

THANK YOU!